

The Financial market is **uncertain** and **non-stationary**; to beat it, one way is to consider portfolio diversification by risk-return tradeoff (portfolio theory).

Problem 1

But LLM in the market, which is not tuned to serve the financial market, excels at extracting clear patterns in plain sight, suffers from uncertainty and non-stationary properties. Because:

- It is trained on large dataset -> regresses toward the means, while holding implicit views (Blackbox nature)
- It is engineered to serve broad group of customers -> quite conservative, better than random portfolio but still very low risk decisions.

The work explores the available LLMs' internal knowledge bias towards:

- implicit view when it comes to making decisions in terms of sector, size, momentum.
- stubbornness when countered with evidence (GPT is quite neutral -> can be implemented with proper RAG).

Choice of LLMs for financial works matters

Problem 2

Using LLMs to embed ETFs needs attention. Solution proposed: "THEME Framework" for fine-tuning LLMs for portfolio selection.

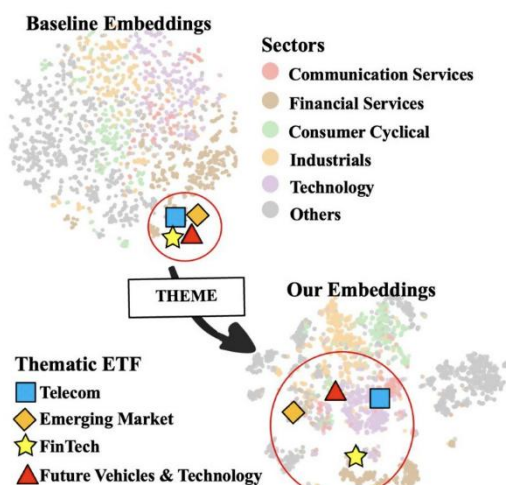
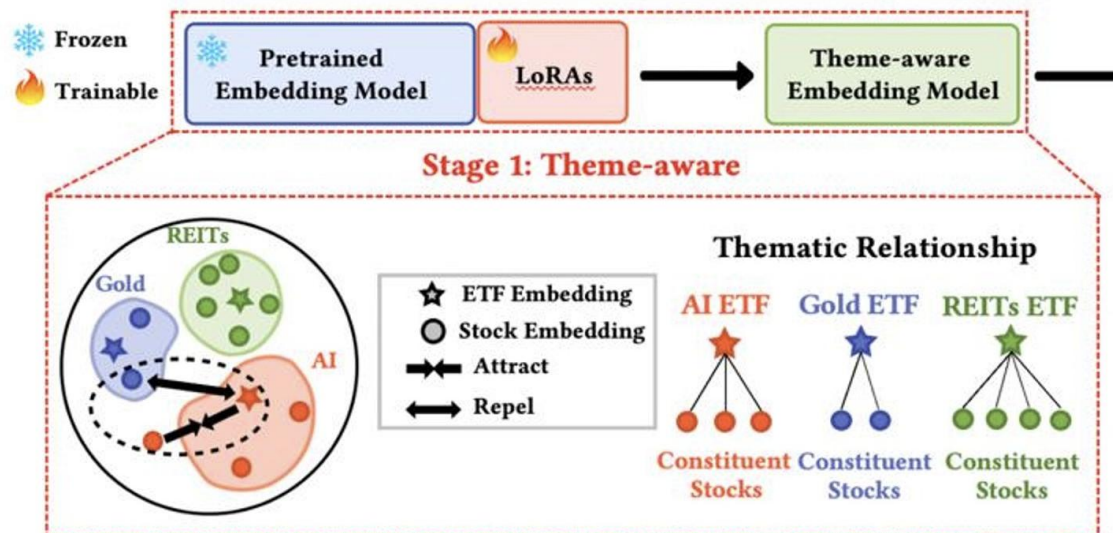
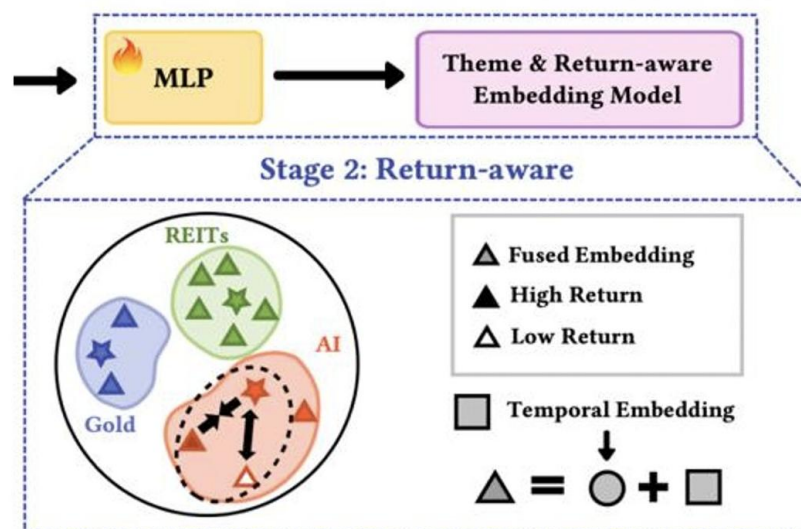


Figure 1: Comparison of t-SNE visualizations of stock description embeddings.

1. Step 1 Theme Awareness: learn semantic embedding space with contrastive learning based on ETF constituent information.

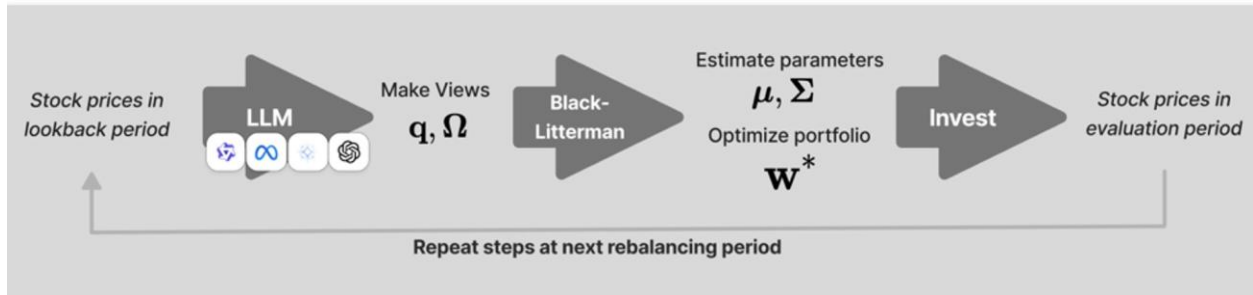


2. Step 2 Temporal Awareness: learn temporal information via contrastive learning based on learned feature from a return prediction model

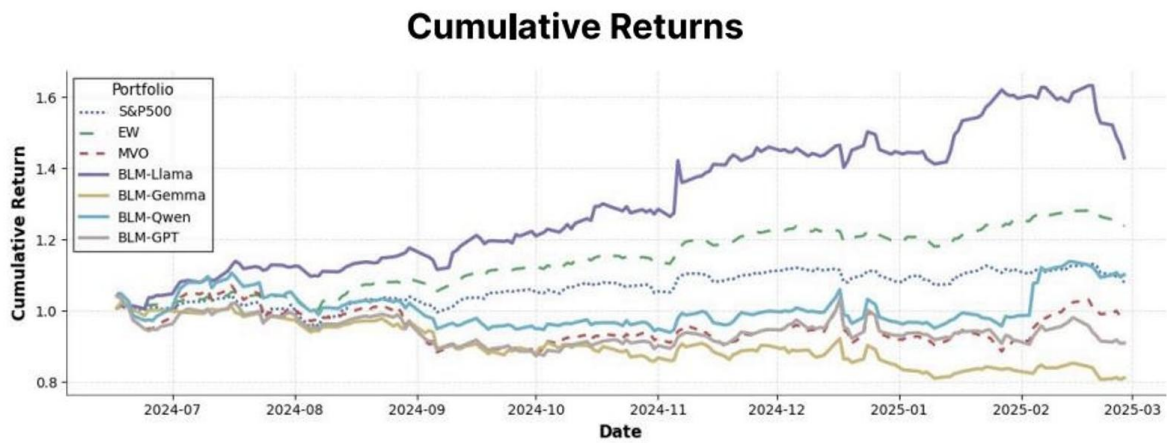


Results: beat pure LLMs and rule-based models in retrieval precision and show superior short-term returns -> crucial for timely rebalancing portfolio.

From problem 1 and 2, they incorporate LLMs into MVO by Black-Litterman model



The experiment result speaks for itself:



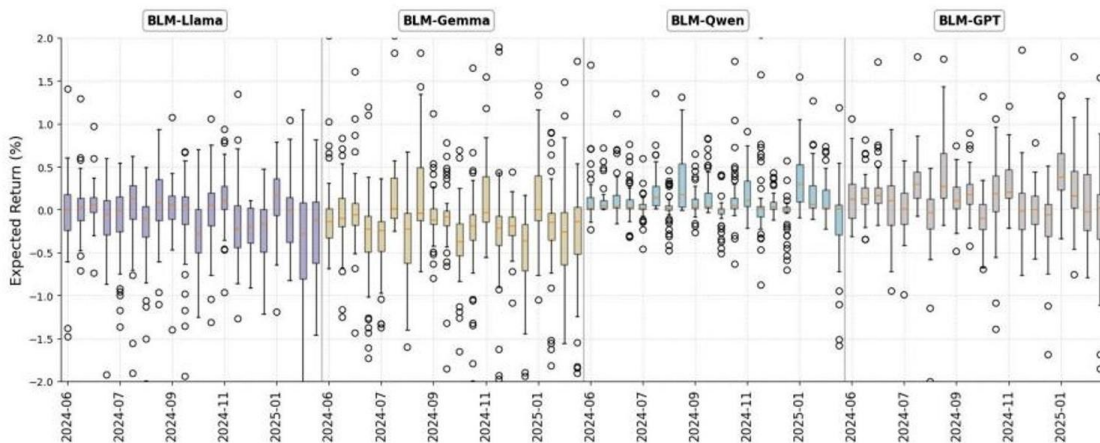
- **BLM-Llama** achieved the highest cumulative return (CAGR: 67.31%)
- **BLM-Gemma, BLM-GPT** delivered negative cumulative returns (CAGR: -0.2593, -0.1276)

Out-of-sample Investment Performance

Metric	S&P500	EW	MVO	BLM-Llama	BLM-Gemma	BLM-Qwen	BLM-GPT
CAGR ↑	0.1159	<u>0.3595</u>	-0.0189	0.6731	-0.2593	0.1494	-0.1276
mean ↑	0.0005	<u>0.0012</u>	0.0001	0.0022	-0.0011	0.0007	-0.0004
std ↓	<u>0.0086</u>	0.0074	0.0182	0.0151	0.0128	0.0160	0.0157
Sharpe ↑	0.0455	0.1580	0.0006	<u>0.1374</u>	-0.0931	0.0375	-0.0318
mean (ann.) ↑	0.1191	<u>0.3142</u>	0.0225	0.5432	-0.2795	0.1710	-0.1056
std (ann.) ↓	<u>0.1371</u>	0.1173	0.2882	0.2398	0.2026	0.2534	0.2486
Sharpe (ann.) ↑	0.7228	2.5075	0.0087	<u>2.1817</u>	-1.4778	0.5959	-0.5054
MDD ↑	<u>-0.0849</u>	-0.0519	-0.1828	-0.1245	-0.2124	-0.1509	-0.1157
VaR95% ↑	<u>-0.0155</u>	-0.0116	-0.0285	-0.0201	-0.0203	-0.0215	-0.0251
CVaR95% ↑	<u>-0.0212</u>	-0.0164	-0.0480	-0.0305	-0.0327	-0.0355	-0.0392

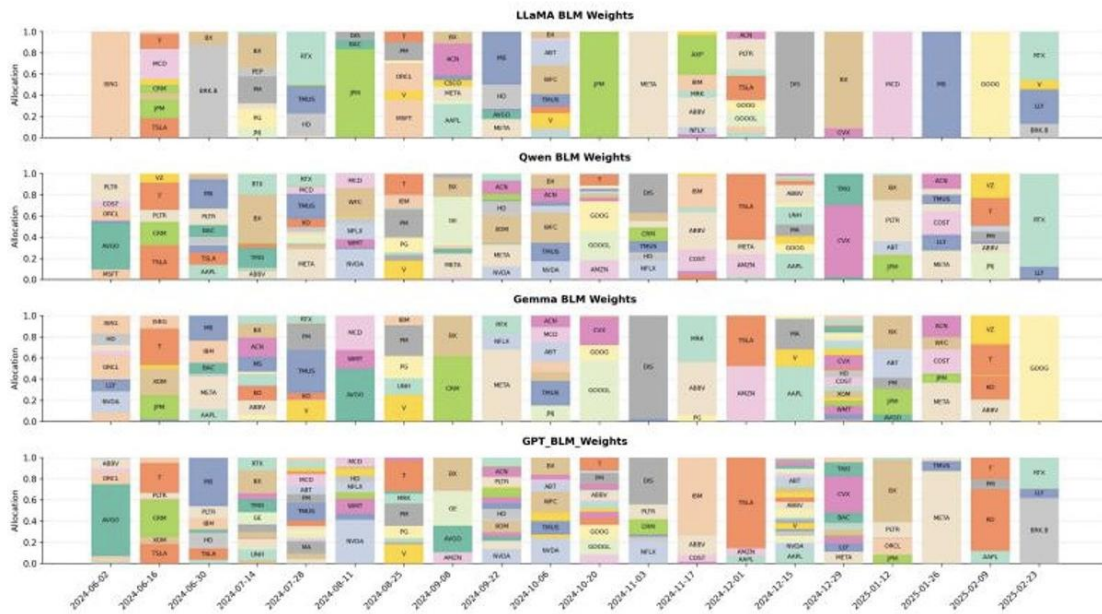
- **EW** recorded the lowest volatility (std: 0.0074)
 - serving as a strong low-risk benchmark
- **MVO** had weakest performance overall (CAGR: -1.89%, Sharpe: 0.0009)
 - suffering from estimation errors and unstable portfolios

Distribution of LLM-generated Views



- **Llama** produced more extreme outliers, especially highly negative views
- **Gemma** showed high variability and a downward bias in predictions
- **Qwen** exhibited neutral and conservative behavior

Time-series visualization of asset allocations



- **BLM-Llama** had concentrated portfolios by strongly filtering out low-potential stocks

Key Takeaways:

1. LLMs have different views and behavior, choose wisely.
2. LLM embedding for investment assets should be different from financial documents (more fine-tuned for them and temporal info).
3. LLMs can be use for portfolio optimization, but don't forget that it's bad with number (gives them tools may be?).